

# Northern Toussian

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Northern Toussian (ISO: tsp; Glottocode: nort2787) is a minority Niger-Congo language spoken in southwest Burkina Faso by the Toussian people. The Toussian languages are of uncertain genetic affiliation—they were originally classified as Gur (Mabia) languages, but the classification of the family has been disputed (Prost 1964, Naden 1989, Mieke 2012, Güldemann 2018). While there is adequate evidence to consider the Toussian languages to be Niger Congo, there has not been sufficient research to confidently assign it to a particular branch of the Niger-Congo family tree. This is much the same situation of the other so-called “peripheral Gur” languages like the Senoufo, Tiéfo, and Samu families, as well as the language Viemo. More precise classification of these languages not only requires historical linguistic work on Gur and each of these groups individually, but also consideration of the relationship of certain members of the Adamawa family, some of which have been shown to share many features in common with Gur languages (Kleinewillinghöfer 1996).

The Toussian languages are spoken in the Cascades and Houets regions of Burkina Faso, west of Bobo-Dioulasso and north of Banfora. The two recognized Toussian languages, Northern Toussian and Southern Toussian, were each reported to have had around 20,000 speakers in 1995 (*SIL Survey Report* 1995). The population of Burkina Faso has roughly doubled since then. Because most varieties of Toussian appear to be vital and are being transmitted to children, we might expect that Northern and Southern Toussian each have around 40,000 speakers currently. In addition to these two Toussian languages, there is a divergent variety spoken only in the villages of Moami and Tien that might constitute its own language—the language of Moami is not mutually intelligible with Northern or Southern Toussian. Preliminary research suggests that it is more similar structurally and lexically to Southern Toussian than Northern Toussian. A map of the Toussian languages and their neighbors is given in Figure 1.

There is considerable dialectal variation in Northern Toussian, with notable differences from village to village. At least three major dialect groups have been identified: the Djigouera, Kourinion, and Guéna dialects (Sawadogo 1995). Prost (1964) considers Guéna-Kourinion to be a single dialect, however these two varieties appear quite distinct—enough so to be considered different dialects.<sup>1</sup> Nonetheless, there has been little

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<sup>1</sup>Sawadogo (1995) performed a mutual intelligibility test whereby speakers of different Toussian varieties were tested on their comprehension of texts of other varieties. It was found that there was a higher degree of mutual comprehension for speakers from Guéna and Djigouera than there was between speakers of either Djigouera and Kourinion or Guéna and Kourinion, suggesting that, if any pair of these three

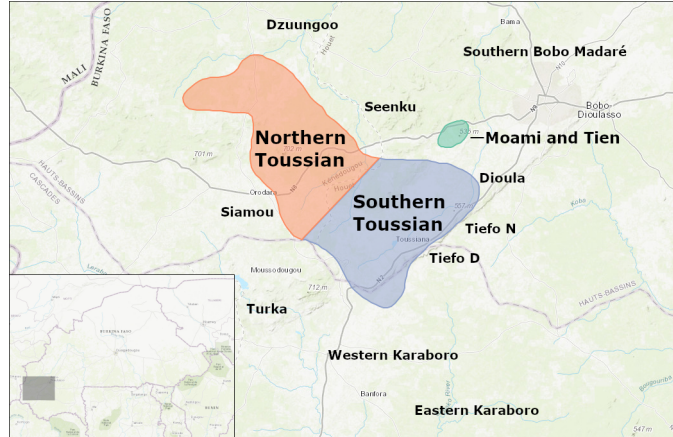


Figure 1: Map of the Toussian languages

dialectological work on Northern Toussian, and as such the number of dialects and their properties remain unclear.

Work on Northern Toussian has only begun in earnest in recent decades. Prost (1964) is a sketch of Southern Toussian, in which he briefly notes some of the properties of the Guéna dialect of Northern Toussian. Some decades later, Zaugg-Coretti (2005) wrote a description of the nominal morphology of Northern Toussian, with data from the Guéna dialect. Recent work includes a phonological sketch of the Kourinion dialect (Bossers & Boone 2023), and a handful of studies based on data from the Djigouera dialect, including a description of the Northern Toussian xylophone surrogate language (Struthers-Young 2022a), a comparison of the nominal morphology of Northern and Southern Toussian (Struthers-Young 2022b), an analysis of double downstep and tonal representation (Struthers-Young 2025b), and a reference grammar (Struthers-Young 2025a, submitted).

This illustration is of the Djigouera dialect, primarily based on data from a 34-year-old man, Daouda Traoré. The data were collected in the field with an AudioTechnica AT875R microphone, recorded on a Zoom F3 field recorder. Some instances of interspeaker variation are shown throughout with data from Karim Traoré, who was 25 at the time the recordings were made. They were recorded with the same microphone, but on a Tascam DR-40X. Unless otherwise noted, the figures throughout were produced based on Daouda's tokens.

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varieties constitutes a single dialect, it would be the Djigouera and Guéna dialects.

## Consonants

|                     | Bilabial | Labiodental | Alveolar | Palatal | Velar | Labialized velar |
|---------------------|----------|-------------|----------|---------|-------|------------------|
| Plosive             | p b      |             | t d      | ɟ       | k     | (ɡb)             |
| Nasal               | m        |             | n        | ɲ       | ŋ     |                  |
| Trill               |          |             | r        |         |       |                  |
| Fricative           |          | f           | s        |         |       |                  |
| Approximant         |          |             |          | j       |       | w                |
| Lateral Approximant |          |             | l        |         |       |                  |

| Consonant | Transcription | Gloss             |
|-----------|---------------|-------------------|
| /p/       | <i>pē</i>     | ‘husband’         |
| /t/       | <i>tēˈ</i>    | ‘take’            |
| /k/       | <i>kēˈ</i>    | ‘surpass’         |
| /b/       | <i>bē</i>     | ‘split’           |
| /d/       | <i>dēˈ</i>    | ‘climb’           |
| /ɟ/       | <i>jē</i>     | ‘sweep’           |
| /ɡb/      | <i>gbátá</i>  | ‘shed’            |
| /m/       | <i>mè</i>     | ‘younger brother’ |
| /n/       | <i>nè</i>     | ‘thin (V)’        |
| /ɲ/       | <i>ɲè</i>     | ‘root’            |
| /ŋ/       | <i>sóŋ</i>    | ‘sting’           |
| /r/       | <i>swār</i>   | ‘termite’         |
| /f/       | <i>fēˈ</i>    | ‘stomach’         |
| /s/       | <i>sē</i>     | ‘chop’            |
| /j/       | <i>jí</i>     | ‘dance’           |
| /w/       | <i>wēˈ</i>    | ‘store’           |
| /l/       | <i>lè</i>     | ‘uncle’           |

There are a number of asymmetries in the consonant inventory of Northern Tossian. A voicing contrast in obstruents is only found with bilabial and alveolar stops. Of the remaining obstruents, some are voiceless, such as /f/, /s/, and /k/; the remaining consonants /ɟ/ and /ɡb/ are voiced. The labial-velar /ɡb/ only occurs in borrowings from Dioula, a trade language that is spoken as L2 by the vast majority of people from southwest Burkina Faso. Certain phonemes are only contrastive in very restricted environments, including the sounds /l/ and /r/, which only contrast as the second sound in a consonant cluster, or /n/ and /ɲ/, which only contrast following /i/. There are a handful of other phones produced as a result of various phonological processes, including [t̪] and [ʃ], palatalized allophones of /t/ and /s/, respectively; [χ]~[x]~[h] and [ɣ]~[ɣ̃], lenited allophones of coda /k/; [ɥ], produced by the sequence /wj/; and [w̃] and [j̃], nasalized allophones of /w/ and /j/. /r/ is pronounced as both [r] and [r̃], with the [r̃] found in slower and more careful speech. The conditioning environments of each of these allophones is discussed below.

## Plosives

Northern Toussian has seven stop phonemes: /b d j gb p t k/. Of these, /gb/ only occurs in Dioula borrowings. A voicing contrast only exists for bilabial and alveolar stops—palatal /j/ is voiced and velar /k/ is voiceless. Phonetically, voiceless plosives are unaspirated. Of the voiceless stops, /p/ has the shortest average VOT at 22.90ms, followed by /t/ at 28.27ms, and /k/ at 36.79ms. The voiced stops are strongly voiced. /b/, /d/, and /gb/ have similar average VOTs at -69.71ms, -75.89ms, and -62.32ms, respectively. /j/ has a slightly shorter average VOT at -43.36ms, with some tokens produced with positive VOTs. The VOTs of each phonemic stop category are plotted in Figure 2, with the average VOT and standard deviation of each category provided in Table 1.

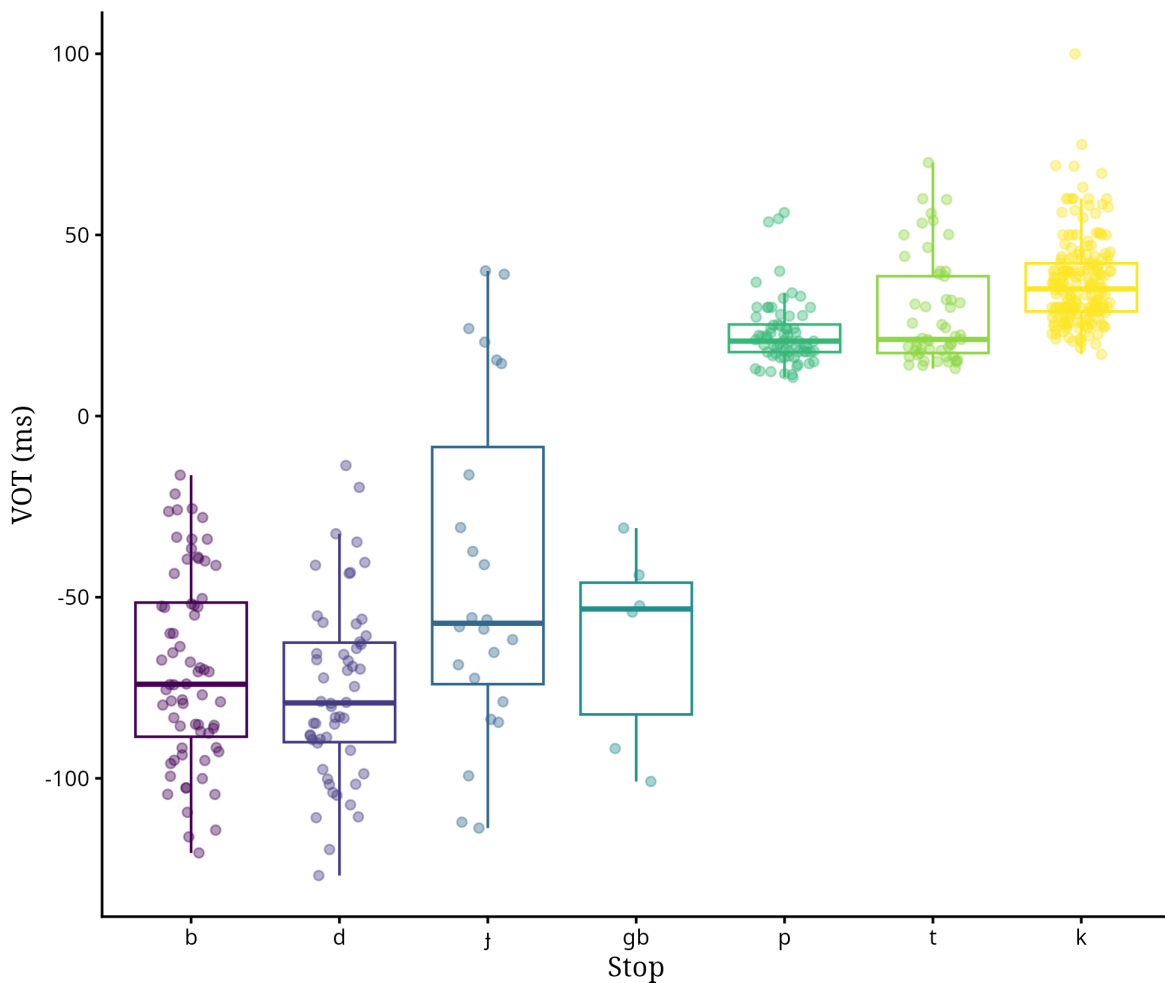


Figure 2: VOT values of each stop category

Table 1: Average VOT by stop category

| Stop | Mean VOT (ms) | VOT sd (ms) |
|------|---------------|-------------|
| /b/  | -69.71        | 26.34       |
| /d/  | -75.89        | 24.62       |
| /j/  | -43.36        | 46.90       |
| /gb/ | -62.32        | 27.73       |
| /p/  | 22.90         | 9.26        |
| /t/  | 28.27         | 14.80       |
| /k/  | 36.79         | 11.86       |

In general, stops undergo little allophony. The single notable allophonic process in stops is that /t/ is palatalized to [tʃ] when followed by the glide /j/. This process is discussed more in the Fricatives section, where it is treated alongside palatalization of /s/.

## Fricatives

There are two phonemic fricatives: voiceless labiodental /f/ and alveolar /s/. The averaged spectra of these sounds are given in Figure 3.

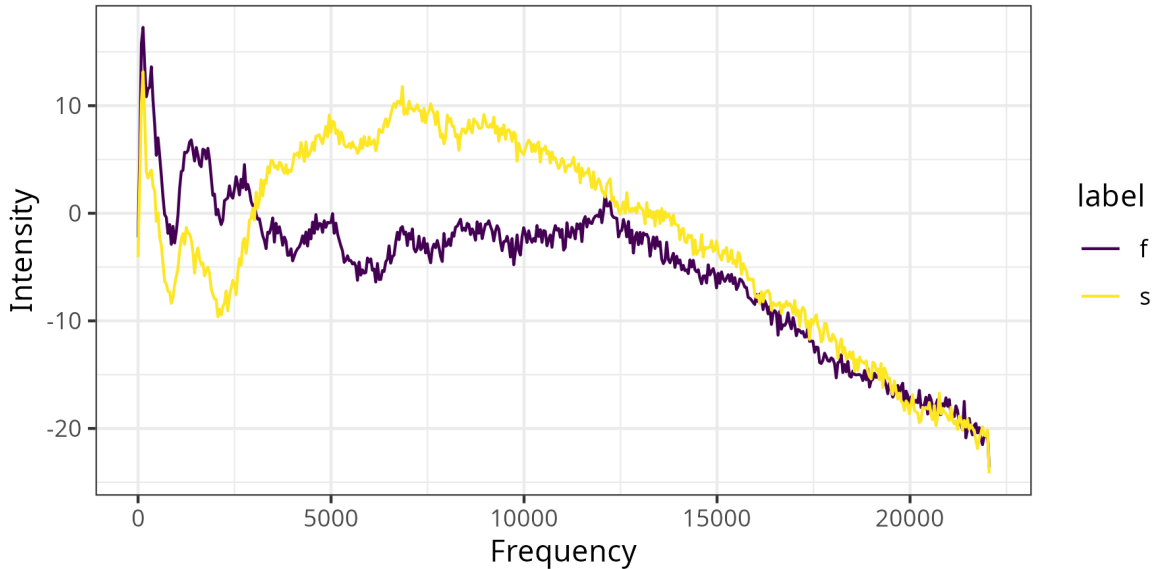


Figure 3: Spectra of [f] and [s]

There are many additional fricative allophones, including [v], [z], [ʃ], [ʒ], and the lenited allophones of coda /k/. [ʃ] is an allophone of /s/, produced as a result of palatalization. [v] and [z] are voiced allophones of /f/ and /s/ that occur after syllabic nasals; [ʒ] occurs when palatalization and postnasal voicing both target /s/. [χ]~[x]~[h] are phrase-final allophones of coda /k/—when coda /k/ is phrase-medial, it is realized as a

voiced fricative [ɣ]~[ʝ] or approximant [ɣ]~[ʝ]. Each of these processes is discussed in the subsequent sections.

## Palatalization

The voiceless alveolar obstruents /t/ and /s/ are palatalized to postalveolar [tʃ] and [ʃ], respectively. The triggering environment varies by dialect: in the Djigouera dialect, /s/ is palatalized by /i/, /j/, and /w/, whereas /t/ is only palatalized by /j/; in the Kourinion dialect, /t/ is palatalized by both /i/ and /j/ (personal research). There is some variation in how regularly palatalization occurs in the Djigouera dialect—/s/ is almost always palatalized when followed by /j/, but the sequence *si* freely varies between [si] and [ʃi]. Typically, when the sequence *sj* is palatalized, the glide /j/ is deleted, meaning the words *sjār* ‘honey’ and *sjō* ‘women’ are generally pronounced as [ʃār] and [ʃō], respectively. The effects of the palatalization can be seen in the spectrograms below. The energy of the frication of [ʃ] and [tʃ] in Figure 4 begins to concentrate around F3-F4, whereas the [s] in Figure 5 begins to increase in energy above F4. These are the expected behaviors of [ʃ] and [s] (Stevens 2000).

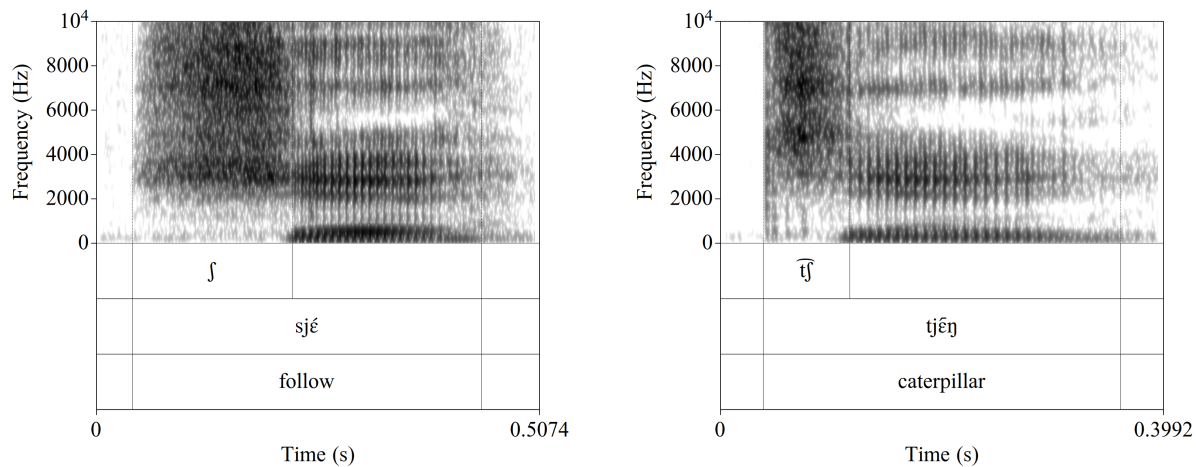


Figure 4: Frication Energy of [ʃ] and [tʃ]

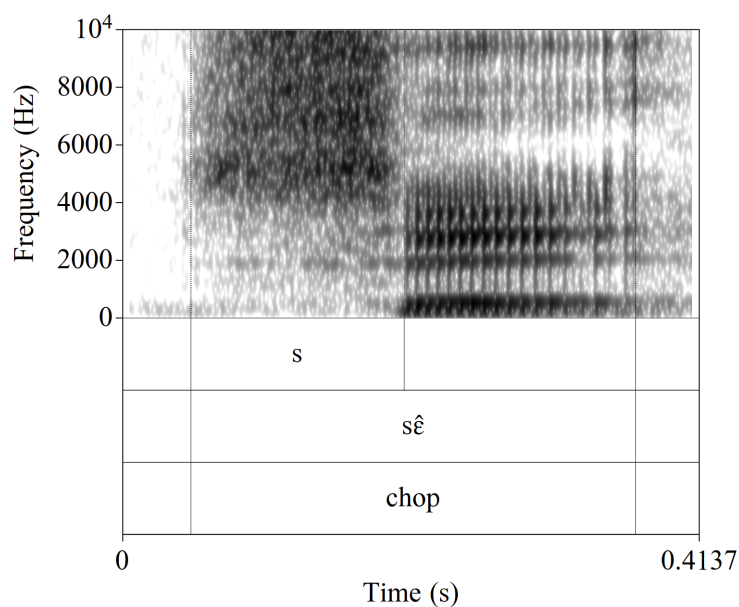


Figure 5: Friction Energy of [s]

Spectra of [s] and [ʃ] are shown in Figure 6.

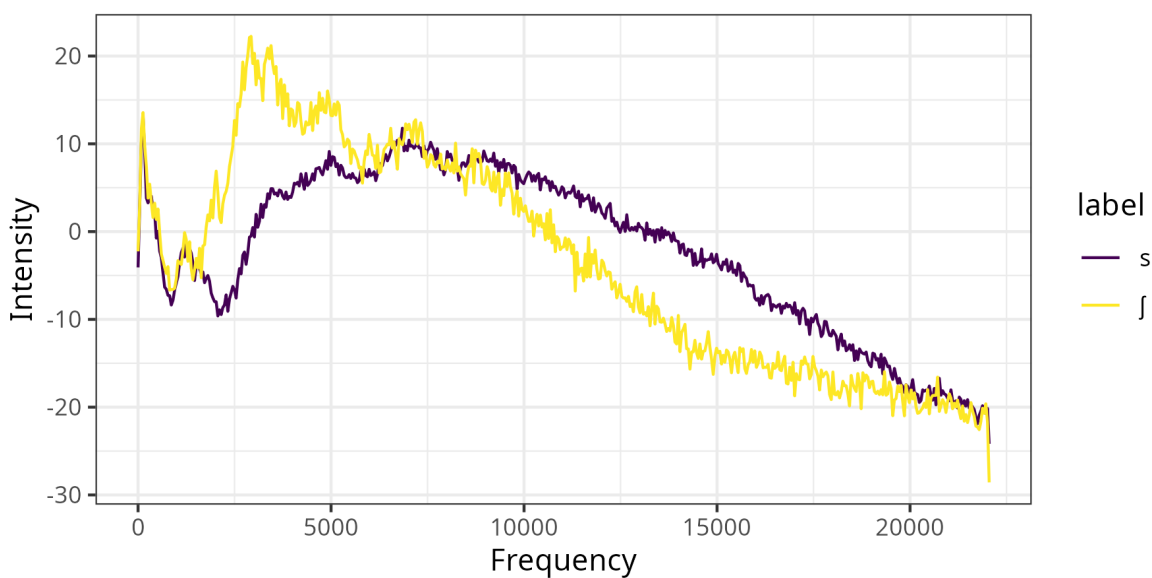


Figure 6: Spectra of [s] and [ʃ]

Given that there is productive palatalization of voiceless obstruents, one might question whether the sound [ʃ] is a palatalized allophone of *d* occurring when *d* precedes *i* or *j*. This is not the case. *ʃ* is a distinct phoneme from *d*, as evidenced by segmental minimal pairs like *dʒà* ‘old’ and *ʃá* ‘watch.’

## Postnasal voicing

The phonemic status of [v] and [z] appears to vary by speaker. With the majority of speakers I have worked with, [v] and [z] are allophones of /f/ and /s/ that occur following syllabic nasals. For some speakers, however, they might constitute marginal phonemes.

Figure 7 demonstrates postnasal voicing. The word *ḥfɪŋ* ‘lie’ is realized as [ḥvɪŋ], and *ḥsɔnɔ̃* ‘kora’ is produced as [ḥzɔ̃nɔ̃].

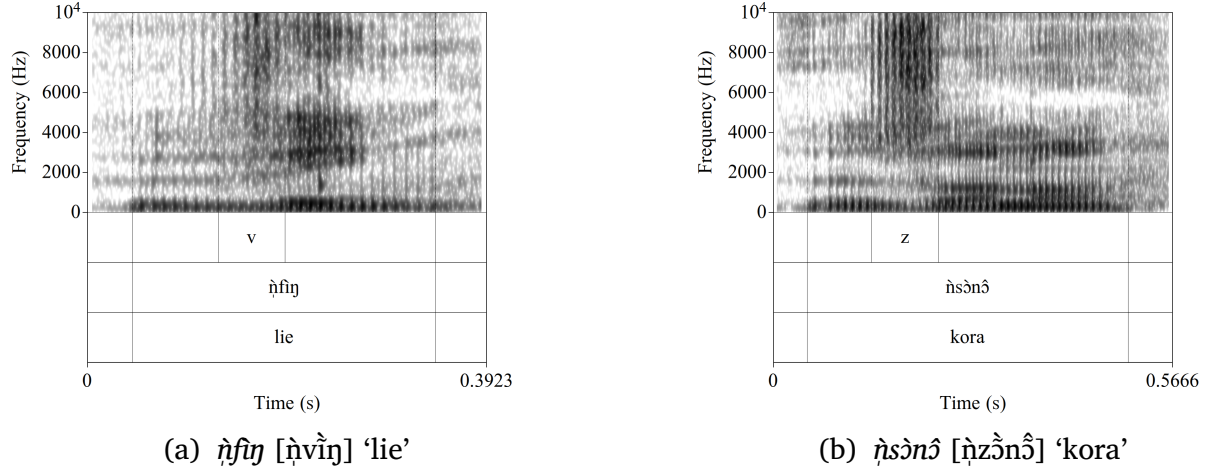


Figure 7: Postnasal voicing

The voicing is only consistently triggered by syllabic nasals, though there appears to be some variability following nasal codas as in the words like *fɔŋsɪ* ‘accept.’ Voicing sometimes occurs with such words, though typically only at faster rates of speech. *s* and *f* are never voiced if they follow a nasal vowel, such as in the word *fɔ̃fɔ̃ŋ* ‘kidney.’ Other examples of voicing triggered by syllabic nasals is shown in Table 2 below.

| Phonemic representation | Phonetic representation | gloss    |
|-------------------------|-------------------------|----------|
| <i>ḥfrɪ</i>             | [ḥvrɪ]                  | ‘soap’   |
| <i>ḥsàk</i>             | [ḥzàχ]                  | ‘pestle’ |
| <i>ḥsɔŋ</i>             | [ḥzɔ̃ŋ]                 | ‘ashes’  |

Table 2: Postnasal voicing

When the sequence *sj* follows a syllabic nasal, both palatalization and postnasal voicing occurs, meaning that words such as *ḥsjɛ̃* ‘wood,’ is pronounced [ḥzɛ̃]. This is shown in the Figure 8.

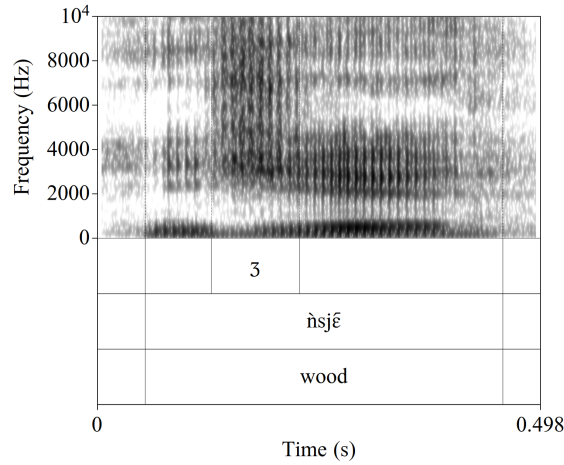
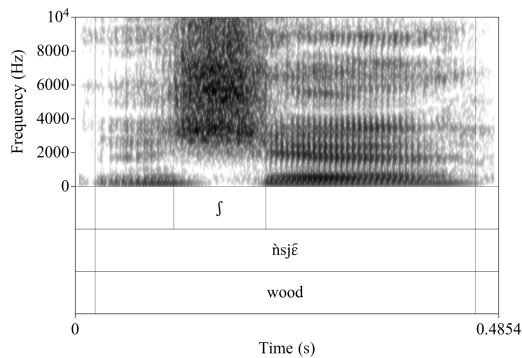
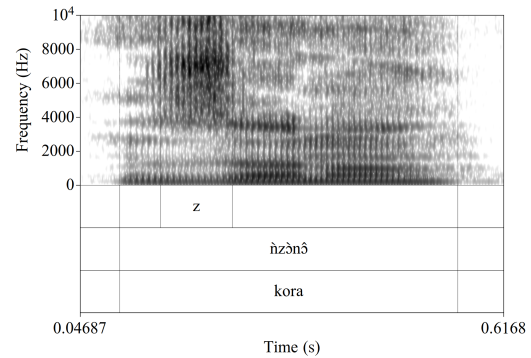


Figure 8: *ɲsjɛ* [ɲʒɛ] ‘wood’ spoken by Daouda

There is interspeaker variation in the application of the postnasal voicing rule—some speakers do not consistently voice *f* and *s* after nasals. One speaker with whom I have worked, Karim Traoré, maintains an apparent contrast between *s* and *z*, with the consonantal minimal pair *ɲsɔ̀nɔ́* ‘tomorrow’ and *ɲzɔ̀nɔ́* ‘kora.’ Daouda, the speaker whose data is the basis for this illustration, applies the voicing rule and pronounces ‘tomorrow’ as [ɲʒɔ̀nɔ́], where the /s/ is realized as voiced [z]—*ɲsɔ̀nɔ́* and *ɲsɔ̀ɲnɔ́* are both valid pronunciations of the word. Figure 9a demonstrates the lack of postnasal voicing in Karim’s idiolect, where *ɲsjɛ* ‘wood’ is produced as [ɲʃɛ], rather than the [ɲʒɛ], as is the case for Daouda in Figure 8. The word ‘kora,’ however is produced as [ɲzɔ̀nɔ́]. This contrast suggests that there might be a marginal /z/ phoneme in Karim’s idiolect. It is unclear at the present moment how commonly /z/ and /v/ are phonemic in the wider community.



(a) *ɲsjɛ* [ɲʃɛ] ‘wood’ produced by Karim Traoré



(b) *ɲzɔ̀nɔ́* [ɲzɔ̀nɔ́] ‘kora’ produced by Karim Traoré

Figure 9: Marginal /z/

## Realization of coda /k/

/k/ is the sole obstruent that can act as a coda. In that position, it has multiple unique phonological properties. First, it is always lenited. Phrase-finally, it is typically produced as a voiceless fricative, varying between velar [x], uvular [χ], or glottal [h], or is entirely deleted. It appears that certain speakers have different preferences—the speaker whose tokens were used for the current illustration most typically produced uvular fricatives. Phrase-medially, it is typically produced as a voiced velar/uvular fricative [ɣ]~[ʁ] or approximant [ɣ]~[ʁ], though it is sometimes voiceless in careful speech. Second, it has a very limited distribution: in the Djigouera dialect, it only follows /a/ or /ã/, though in other dialects it can occur in wider contexts. In the Kourinon dialect, it occurs following /ɔ/, /u/, and /ə/, but not /i/ or /ɛ/, or any nasal vowel, including /ã/ (Bossers & Boone 2023). Third, whenever coda /k/ occurs phrase-medially, an epenthetic *a* typically follows the /k/.<sup>2</sup>

Figure 10 shows phrase-final coda /k/, where it is pronounced as [χ].

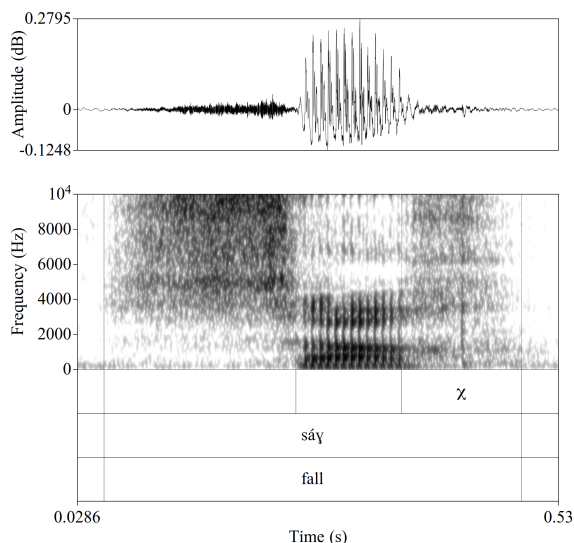


Figure 10: *sák* [sáχ] ‘fall; await’

Figure 11 is phrase-medial coda /k/, pronounced as the approximant [ɣ].

<sup>2</sup>I analyze that phrase-medial [Cakə] sequences are derived from underlying /Cak/—rather than reflecting /Caka/ sequences that undergo phrase-final apocope of the /a/ and lenition of the /k/ to [Cax]—for multiple reasons. First, the epenthetic vowel always copies the immediately preceding tone, e.g., *dāk* ‘way’ becomes [dàkà], *pápák* ‘water fowl’ becomes [pápáká], and *ínák* ‘girl’ becomes [ínákà]. Second, /a/ does not undergo apocope in any other contexts. Third, the morphophonological behavior of the identificational marker, which involves prolongation of the final segment of the word, does not act in the same way as it does with words of the form /CVCV/, suggesting a different underlying form. The first and second points are elaborated in Struthers-Young (2025a: pp. 70–72), and the third in Struthers-Young (2025a: p. 146).

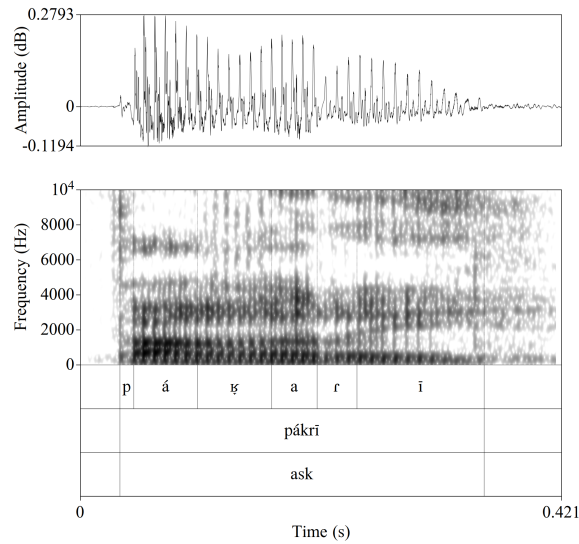


Figure 11: *pákrī* [páɣrī] ‘ask’

The average spectra of [χ] are compared to [f] and [s] in Figure 12.

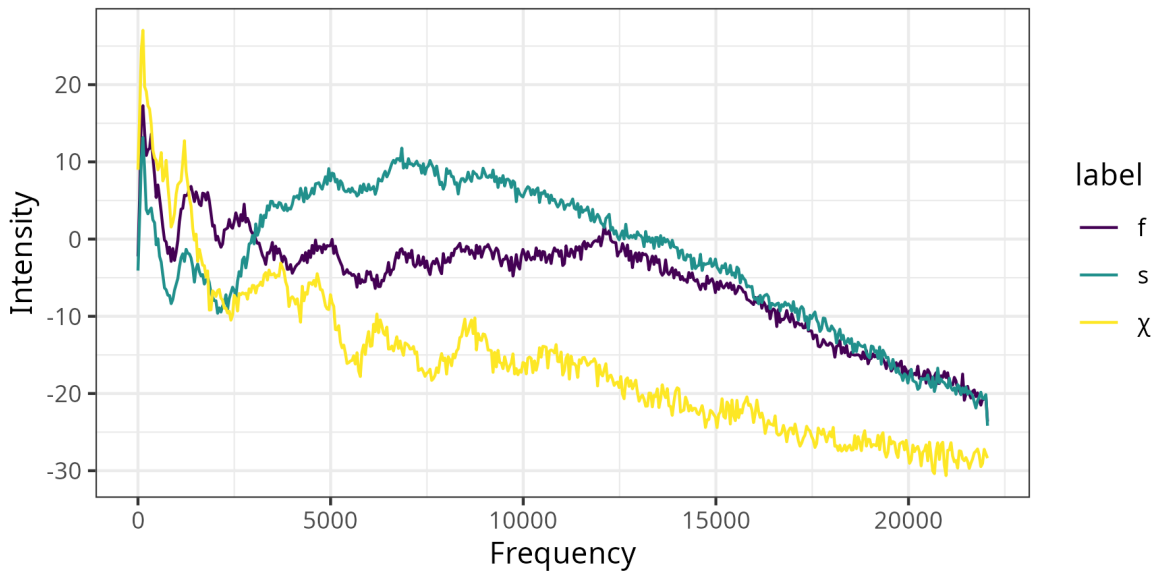


Figure 12: Spectra of [χ] compared to [s] and [f]

## Approximants

Northern Toussian has the following approximants: /l r j w/. /r/ is a rhotic that is realized either as [r] or [ɾ]. The [ɾ] allomorph is generally more common, especially at fluent rates of speech. [r] occurs in careful speech, and is particularly common when following /ə/ or when /r/ is prolonged in the identificational construction (Struthers-Young 2025a:

pp. 70–72). Outside of function words like *ré~ré* ‘at’ or *rí* SBJV, /r/ only occurs as a coda or as the second consonant in an onset cluster like the word *drê* ‘tail’ or *fribú* ‘beetle.’ /l/ can never occur as a coda consonant, and is restricted to the onset, where it can be a singleton consonant or the second consonant in a cluster. This means that the contrast between /l/ and /r/ is limited, only found when /l/ and /r/ are the second sound in a cluster. Indeed, when Dioula words are borrowed into Northern Toussian and nativized, the sounds /l/ and /r/ often alternate. For example, Dioula *kàlîfá* ‘entrust’ is borrowed as *kàrfó* in Northern Toussian; Dioula *kàlâ* ‘study, read’ is attested both as *kàlâ* and as *kâr*—the patterns expected if *l* is restricted to the onset and *r* to the coda. However, these sounds are distinct phonemes, evidenced by minimal pairs like *krô* ‘bambara ground nut’<sup>3</sup> and *klô* ‘tire (V).’

Glides are generally restricted to the onset, where they can be singleton consonants or part of a cluster like *fw* or *tj*. An exception to this is the sequence *ɛj*, which is well-attested in many native words like *pɛj* ‘do’ and *tɛj* ‘partridge.’ There are two notable instances of glide allophony. When glides precede or follow nasal vowels, they are nasalized. This is seen with the words *wê* ‘star’ and *jê* ‘eye,’ which are pronounced as [w̃ê] and [j̃ê], respectively. Additionally, the sequence *wj~wi* is often pronounced as [ɥj]~[ɥ]. This is seen with the word *wi* ‘return,’ pronounced as [ɥi]. [ɥ] is likewise nasalized before nasal vowels.

One of the glides in a *wj* sequence is often deleted. For example, the *j* in *wjê* ‘cord’ is sometimes omitted, and the word is attested as [w̃ê]. Conversely, both *wjí* ‘return’ and *wjú* ‘alive’ are attested without the *w*, pronounced as [jí] and [jú], respectively. There does not appear to be a systematic pattern to account for which glide is deleted; instead there is both inter- and intraspeaker variation. The speaker for this illustration did not delete either glide in such tokens, generally producing the [ɥ] allomorph instead.

## Vowels

Northern Toussian has a large vowel inventory: in addition to eight modal oral vowels, it has six modal nasal vowels /ĩ ẽ ã ũ õ ỹ/, though /ĩ õ ũ/ are marginal.<sup>4</sup> There is also a series of non-modal vowels that have been characterized variously as ‘glottalized’ (Prost 1964) and ‘pharyngeal(ized)’ (Zaugg-Coretti 2005, Bossers & Boone 2023). In the Djigouera dialect of Northern Toussian, there are four non-modal vowel phones, with only two contrastive non-modal vowel phonemes. /ɛ̣/ and /ẹ̃/ are phonemes distinct from /ɛ/ and /ẽ/, but [ạ] and [ạ̃] are allophones of /a/ and /ã/ that occur when /a/ and /ã/ are in open syllables.<sup>5</sup> As I discuss in the Non-modal vowels section below, these vowels have distinct phonetic correlates, where [ạ] has many of the properties of canonical

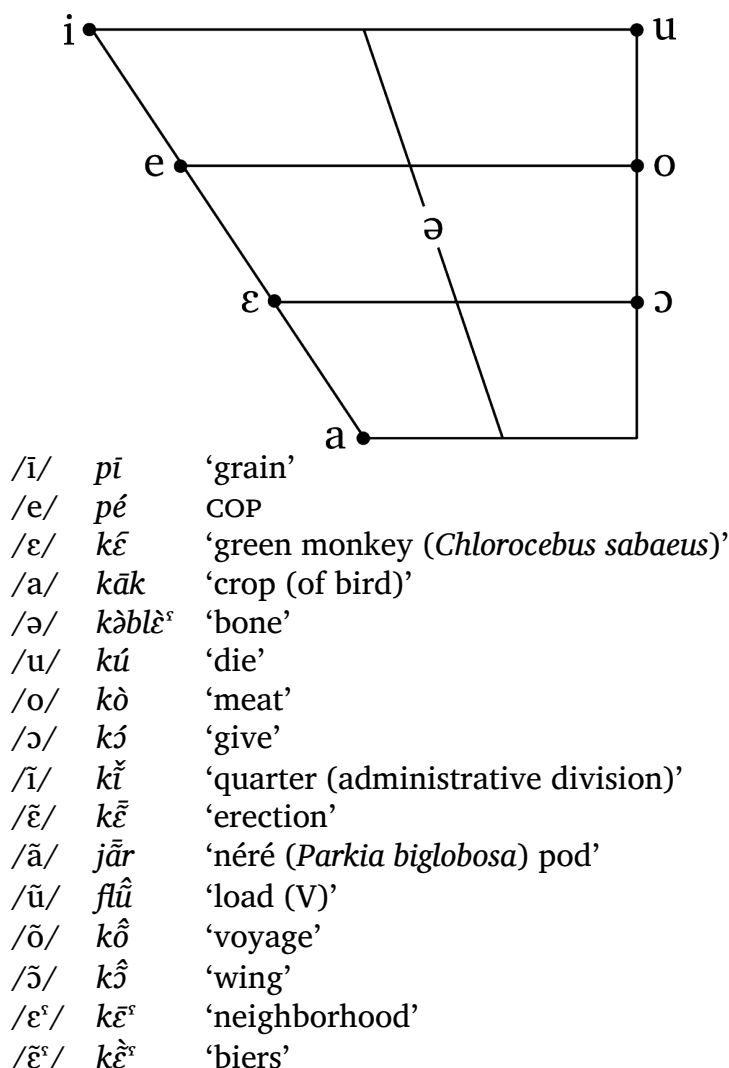
<sup>3</sup>This word might reflect a nativized borrowing from Dioula *tiganinkuru* ‘bambara ground nut,’ indicating that /r/ might have only recently phonemicized.

<sup>4</sup>Some Dioula borrowings use /ẽ/ as well, e.g., *bárdê* ‘slave,’ though this vowel is often adapted to [e] or [ẽ] when borrowed into Northern Toussian.

<sup>5</sup>Bossers & Boone (2023) note that the Kourinion dialect has a contrast between /a/ and /ạ/ as well as /ã/ and /ạ̃/, for which they provide minimal pairs. However, no minimal pairs were found in the Djigouera dialect.

pharyngeal vowels, as they appear creakier and have higher F1 than corresponding [a], whereas [ɛ<sup>s</sup>] is breathier than [ɛ] and has no significant difference in F1.

The vowels of Northern Toussian are shown below:



## Oral vowels

The high mid vowels *e* and *o* are phonetically quite close to the high vowels *i* and *u*. In the Djigouera dialect, the vowel *ə* has a restricted distribution: it cannot occur word finally in lexical items, though it is word-final in some function words like *pə* 3PL.H or *kə* NEG. Most often it occurs in words of the shape CəC(C)V(C). There is considerable variation in the vowel system across dialects. In the Kourinion dialect, *ə* regularly occurs word finally, and it lacks the phonemes /o/ and /e/ (Bossers & Boone 2023). Based on regular correspondences with the Djigouera dialect, /o/ appears to have merged with /ə/ and /e/ with /ɛ/.<sup>6</sup> The merger of /o/ with /ə/ could explain why /ə/ can occur word-finally

<sup>6</sup>C.f., Djigouera *dó* ‘lick’ with Kourinion *dá* ‘lick’ (Bossers & Boone 2023: p. 35), and Djigouera *bē* ‘carry on the back’ with Kourinion *bē* carry (Bossers & Boone 2023: p. 24)

in the Kourinon dialect—as former /o/ centralized to [ə], this led to instances where [ə] occurred word-finally. This then resulted in the loss of the phonotactic restriction.

The F1 and F2 values of these vowels are plotted in Figure 13. Formant values were extracted with VoiceSauce (Shue, Keating & Vicens 2009) from the midpoint of the vowel and estimated with the Snack toolkit (Sjölander 1999). Back vowels, whose F1 and F2 are quite close, were often mismeasured, such that F1 and F2 were conflated and F3 was interpreted as F2, resulting in tokens that appear much more central than is accurate. Any vowels that appeared to be mismeasured were checked for accuracy and hand-measured if they were indeed mismeasured.

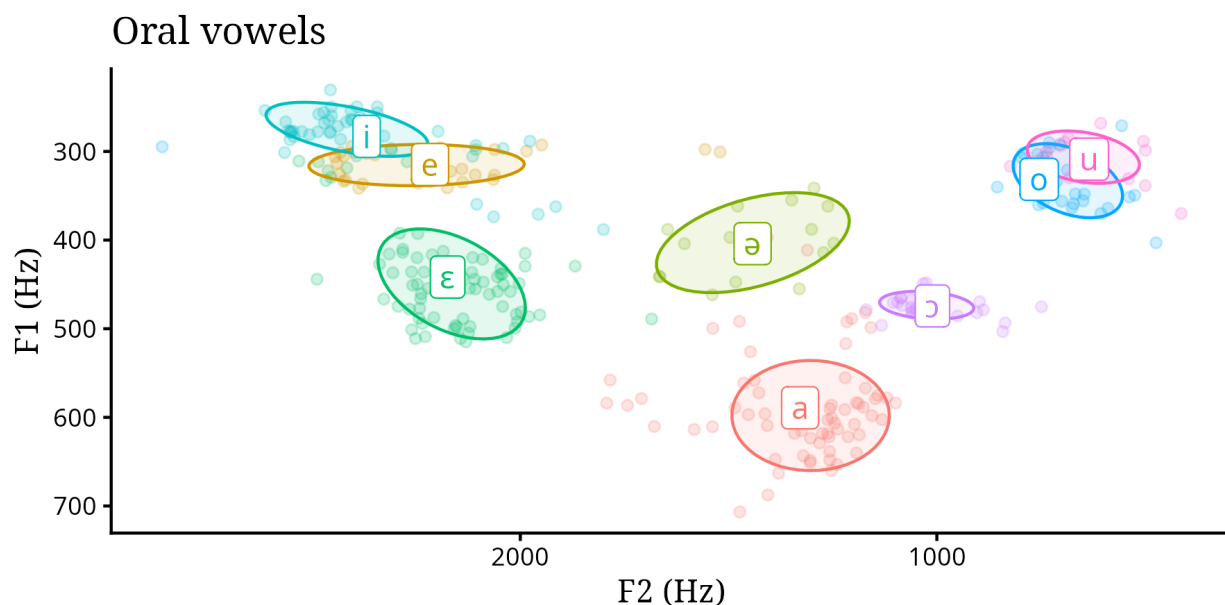


Figure 13: Northern Toussian oral vowels

## Nasal vowels

While Northern Toussian has seven contrastive nasal vowels /ĩ ẽ ã ü õ ỹ/, the vowels /ĩ/, /ũ/, and /õ/ are marginal, only occurring in a handful of tokens. /õ/, in particular, is exceedingly rare—outside of Dioula borrowings, it only occurs with a class of derived verbal nouns like *kô* ‘trip, voyage,’ from *kô* ‘walk,’ *dô* ‘sleep (N)’ from *dó* ‘sleep (V),’ or *klô* ‘fatigue’ from *klô* ‘tire.’ Plain (i.e., modal) nasal vowels are plotted in Figure 14. Because of the dampening effects that the nasal zeros have on F1, VoiceSauce and Praat often mistakenly interpreted P0 as F1, resulting in inaccurate measurements for most tokens. Consequently, each of the vowels in Figure 14 was manually measured in Praat, resulting in a smaller sample of vowels than the oral vowels above.

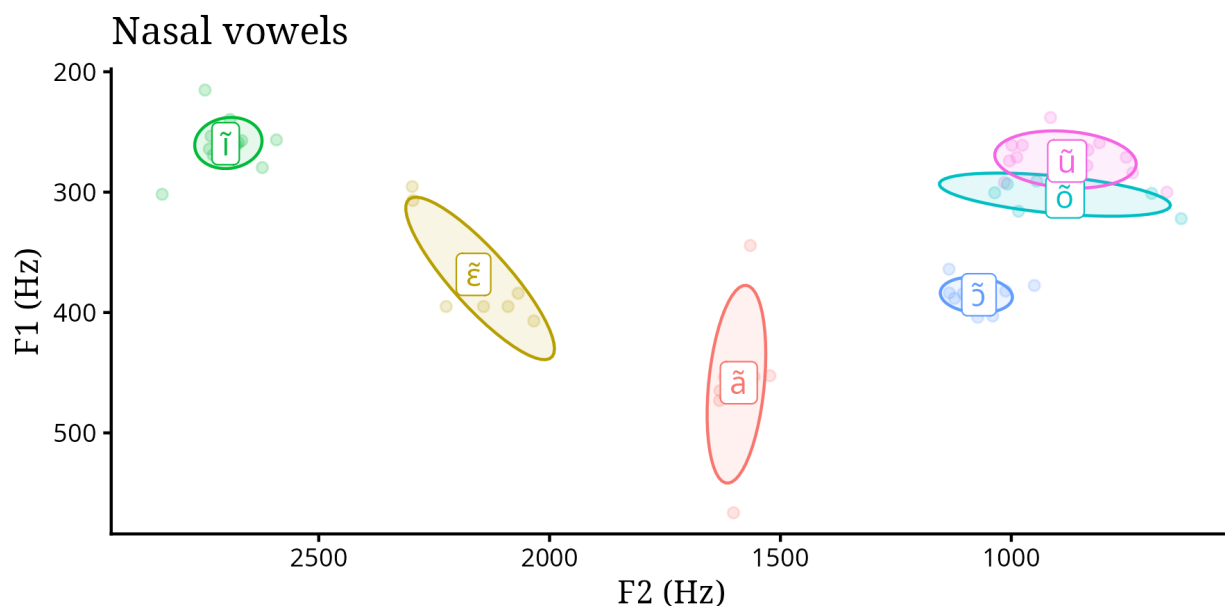


Figure 14: Nasal vowels

Broadly, the nasal vowels have a lower F1 than their corresponding oral vowels and the back vowels are more centralized.<sup>7</sup> Oral vowels are plotted alongside nasal vowels in Figure 15.

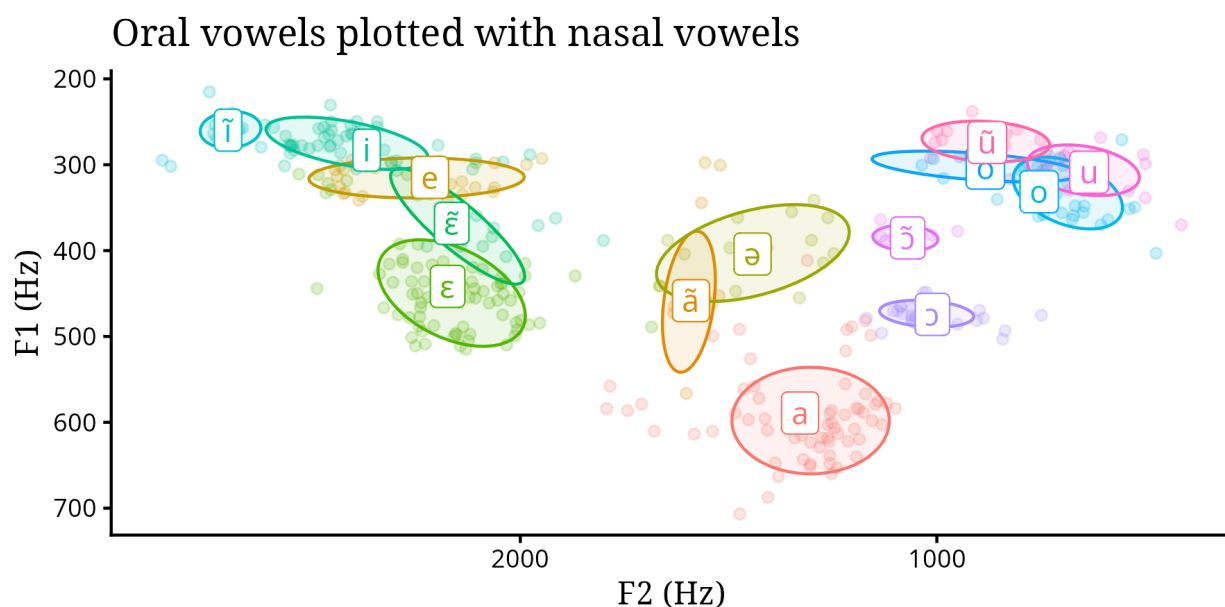


Figure 15: Nasal vowels plotted alongside the oral vowels

The *a* and *ɔ* vowels freely vary between oral or nasal in certain words, such as those in

<sup>7</sup>Some of these differences could be attributed to differences in measuring methods, as each automatic formant tracking algorithm has some degree of inherent error (Schiel & Zitzelsberger 2018).

Table 3. This is not the case for other words like *jā* ‘face,’ or *dò* ‘roan antelope,’ which are only attested with oral vowels, as well as *jǎ* ‘dragonfly,’ which only occurs with a nasal vowel.

| Word           | Gloss     |
|----------------|-----------|
| <i>kɔ̌~kɔ̃</i> | ‘give’    |
| <i>swā~swǎ</i> | ‘field’   |
| <i>kā~kǎ</i>   | ‘kitchen’ |

Table 3: Words with vowels which are variably oral and nasal

Nasal vowels do not contrast with oral vowels before or after tautosyllabic nasal stops. In these contexts, they are always nasalized, such as the word *mè* ‘younger brother,’ pronounced as [mɛ̃], or *kūŋ* ‘funeral,’ realized as [kũ̃ŋ]. For some speakers, ɔ̌ is centralized before ŋ, where it is realized as a nasalized schwa [ə̃], e.g., *kòŋ* ‘buffalo,’ pronounced as [kə̃ŋ].

## Non-modal vowels

All attested varieties of Toussian have a series of contrastive non-modal vowels. These have been classified differently depending on the author: Prost (1964) described the Northern and Southern Toussian non-modal vowels as glottalized, whereas Wiesmann (2004), Zaugg-Coretti (2005), and Bossers & Boone (2023) called these vowels pharyngeal.

Cross-linguistically, pharyngeal vowels are characterized by a higher F1 than their modal counterparts, alongside changes to F2 and F3 (Al-Tamimi 2017). F2 tends to be lower in pharyngeal vowels (Al-Tamimi 2017, Buech, Hermes & Ridouane 2025), though for some languages F2 is higher across the vowel inventory or lower for front vowels and higher for back vowels (Faytak, Quintana Godoy & Yang 2024, Arkhipov et al. 2019). For yet other languages, the changes to F2 vary by speaker and vowel quality (Chiu & Sun 2020). Like F2, the changes in F3 vary widely by language. Some languages exhibit lowering for all vowel qualities, while in other languages, the F3 of some vowels is raised and others is lowered—the vowels that condition raising or lowering is language-specific (Al-Tamimi & Heselwood 2011, Chiu & Sun 2020). As will be seen below, [aʕ] has a higher F1 than its modal counterpart, alongside a lower F3. Formant values of [ɛʕ] are similar to those of [ɛ], meaning that the contrast is likely cued by other phonetic features, such as phonation.

The Djigouera variety Northern Toussian has oral [aʕ] and [ɛʕ] and nasal [ãʕ] and [ẽʕ]. The oral non-modal vowels are shown in Figure 16, plotted alongside the plain oral vowels. As would be expected for pharyngeal vowels, [aʕ] has a higher F1 than [a]. However, there is not much difference of F1 or F2 in [ɛ] and [ɛʕ].

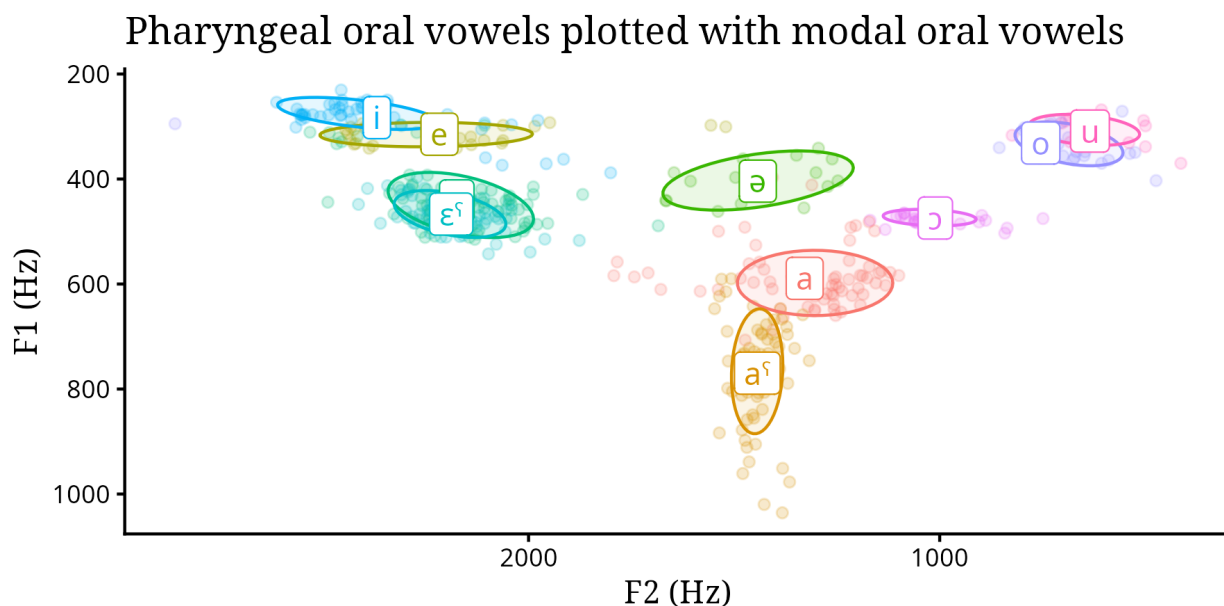


Figure 16: Oral pharyngeal vowels

The F3s of the non-modal vowels tend to be lower than their modal counterpart, as shown in Figure 17, with the difference being greater between [a̠] and [a] than [ε̞] and [ε]. Because of the difficulty in extracting accurate formant values from the non-modal nasal vowels, I only provide a plot for the oral vowels.

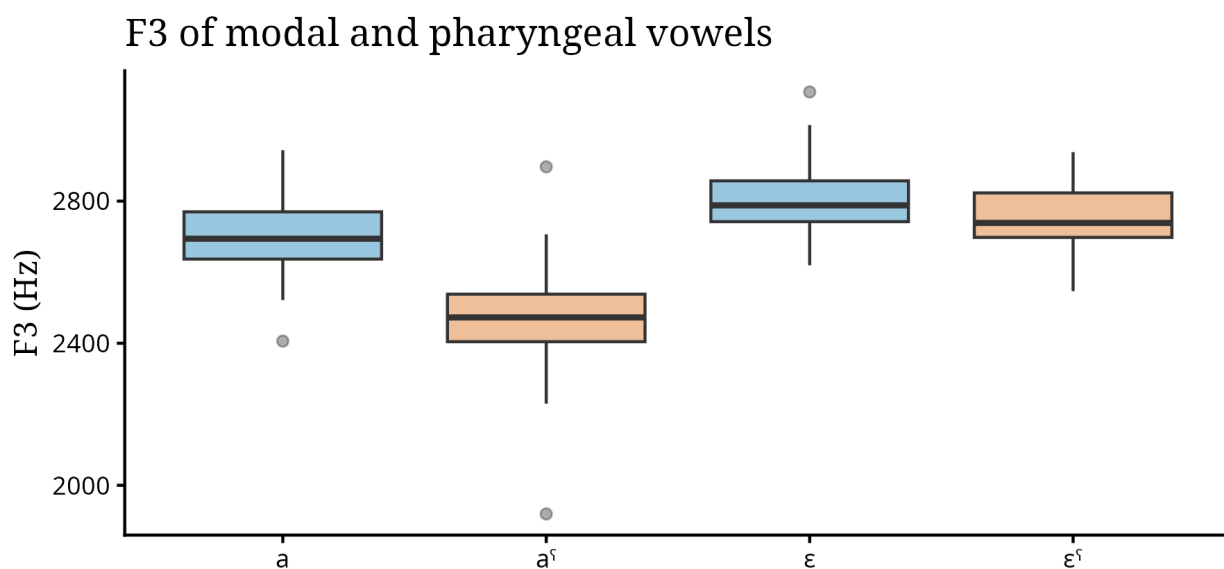


Figure 17: F3s of plain and pharyngeal oral vowels

There are changes to the phonation of these vowels, though the precise nature of changes is as yet unclear—preliminary work suggests that [a̠] is creakier than [a] and

[ɛ̃] is breathier than [ɛ], though further work is needed to determine to what degree position within the utterance and interspeaker variation influences the phonation of these vowels (Struthers-Young & Mamadou Y. submitted). Canonical pharyngeal vowels are expected to be creakier, not breathier (Esling 2005)—this raises questions as to whether these vowels truly are pharyngeal, or might instead have different type of secondary articulation or phonation. Henceforth, I refer to this category of vowel as pharyngeal, with the caveat that more thorough phonetic analysis of these vowels might result in them being reclassified.

The strength of pharyngealization appears to depend in part on discourse factors. The pharyngealization tends to be quite weak in careful speech, and vowels like ɛ̃ and ɛ are perceptually quite similar—this is supported by the similarity of /ɛ/ and /ɛ̃/ in Figures 16 and 17. In fluent and animated speech the pharyngealization is often stronger and more distinct.

## Tone

Northern Toussian has three level tones: a high tone (*á*), mid tone (*ā*), and low tone (*à*). Additionally, it has six lexical contour tones, HM (*ā̃*), HL (*â*), LH (*ǎ*), LHL (*ǎ̃*), LHM (*ǎ̃̃*), and HLH (*ẫ*), though the HLH contour is only attested in one word, *bễ* ‘again.’ I analyze the contour tones as being comprised of two or more of the level tones. Unlike many languages of Southeast Asia where particular tones are associated with different phonation types (Brunelle & Kirby 2016), pharyngealization is orthogonal to tonal assignment in Northern Toussian, i.e., pharyngeal and plain vowels can host all tonal categories. The following table demonstrates the different tonal categories attested in Northern Toussian.

| Tone | Word         | Gloss                                     |
|------|--------------|-------------------------------------------|
| H    | <i>bá</i>    | ‘porridge’                                |
| M    | <i>dā</i>    | ‘shea ( <i>Vitellaria paradoxa</i> ) nut’ |
| L    | <i>bà</i>    | ‘limit’                                   |
| HM   | <i>dē̃</i>   | ‘dream’                                   |
| HL   | <i>dē̃̃</i>  | ‘rock’                                    |
| LH   | <i>bǎ</i>    | ‘poison’                                  |
| HLH  | <i>bễ</i>   | ‘again’                                   |
| LHL  | <i>dǎ̃̃j</i> | ‘wild canid’                              |
| LHM  | <i>jǎ̃̃</i>  | ‘first daughter’                          |

Pitch traces of the three level tones is given in Figure 18. The pitches of the words used to generate these figures were measured at millisecond intervals by VoiceSauce (Shue, Keating & Vicens 2009). These measurements were time-normalized by tonal category, meaning that the length of a given tone in the chart reflects the average length of said tonal category.<sup>8</sup> These data were then averaged. Both the H and M tones are rather

<sup>8</sup>It should be noted that the words were not controlled against frequency effects or the inherent length of vowels, so the any difference in length between tones could be due to reasons other than intrinsic properties of the tones.

level across the syllable. The L is characterized by a slight phrase-final fall. Since most tokens used for this analysis were monosyllabic and spoken in isolation, this fall is readily apparent in the plot.

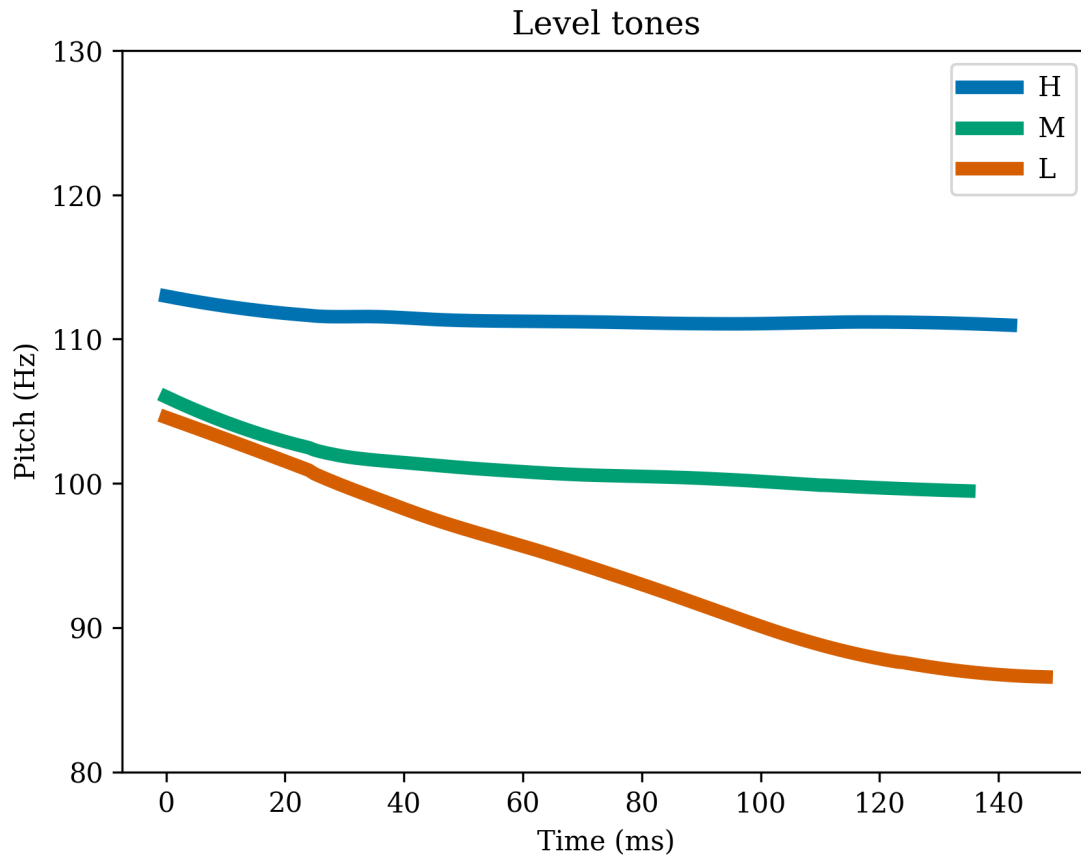


Figure 18: Level tones

Of the six logically possible two-tone contour tones, Northern Toussian only has three: HM, HL, and LH. It is not unusual that there are more falling contour tones than rising contour tones—cross-linguistically, rising contour tones are more marked than falling tones, and tend to have more distributional restrictions (Zhang 2001: pp. 122–133). Pitch traces of two-tone contour tones are given in Figure 19.

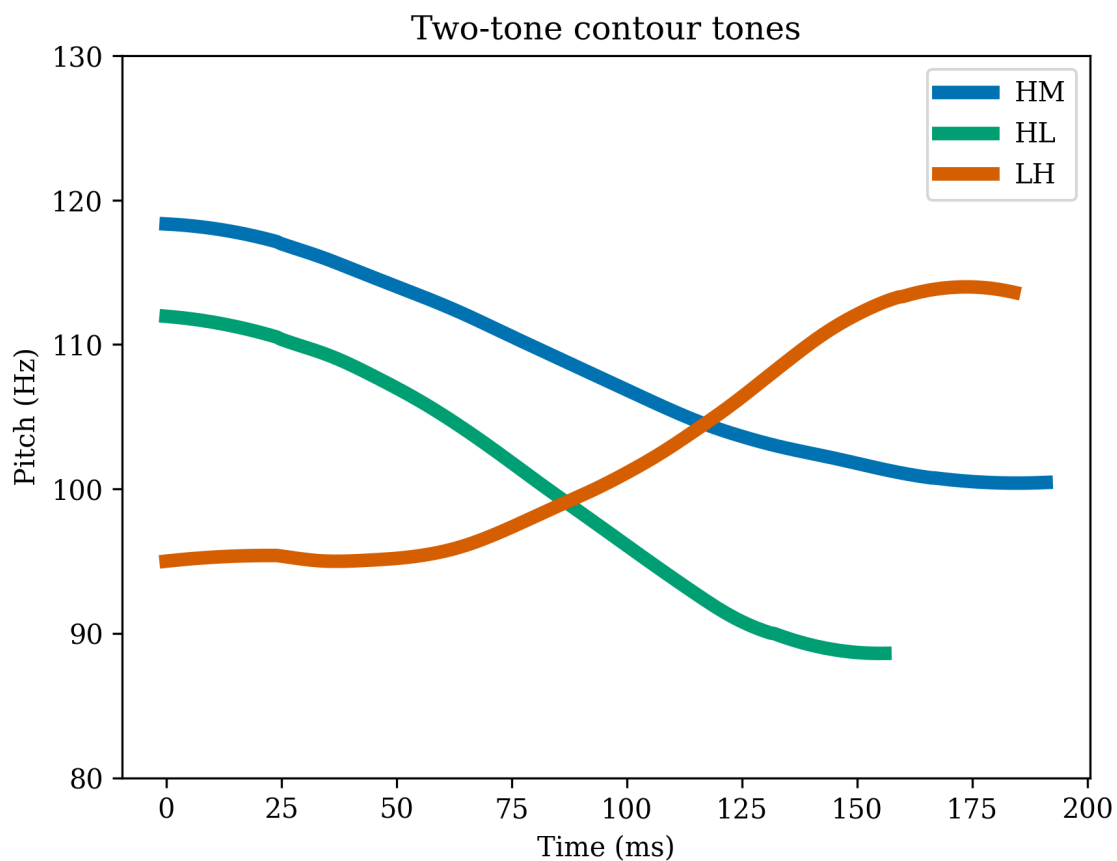


Figure 19: Two-tone contour tones normalized

The level tones and two-tone contour tones are plotted together in Figure 20. The two-tone contour tones are on average longer than the level tones by about 50ms. This is a phonetic effect whereby the syllable is prolonged to accommodate the two tonal targets.

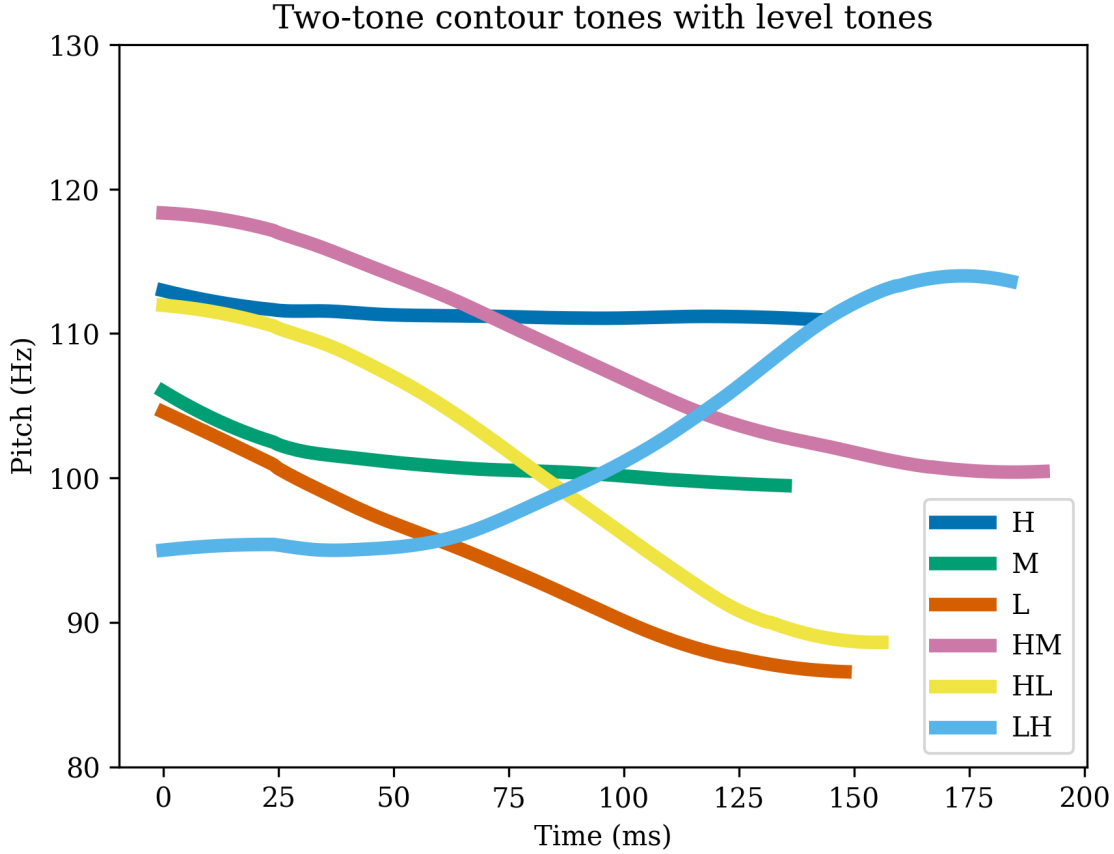


Figure 20: Normalized level tones and two-tone contour tones

In Figure 20, the HM contour tone begins at a higher initial pitch than the H. This led Bossers & Boone (2023) to analyze the tonal category as a contour tone consisting of an extra high tone followed by a high tone (xHH). There are several pieces of (morpho-)phonological evidence that support an HM analysis over an xHH one. First, an xHH analysis requires there to be a phonological extra-high tone, which does not exist in any context other than in this particular contour tone. Second, HM contour triggers several processes that only M tones do. For example, there is an M spreading rule whereby L determiners are realized as M following an M; otherwise, they are realized as L (Struthers-Young 2025a: pp. 150–2). This is demonstrated with the proximal determiner in (1), where it is pronounced with its lexical L following H, HL, or L nouns in (1a), but as *ā* following HM and M nouns in (1b).

- (1) a. H     *bú à* ‘this leopard’  
       HL    *flê à* ‘this woman’  
       L     *dò à* ‘this roan antelope’
- b. M     *bō ā* ‘this eggplant’  
       HM    *ké ā* ‘this green monkey’

There is also a process whereby an M positioned at the right edge of phonological phrases cause following tones to be downstepped. This phenomenon is discussed in detail in Struthers-Young (2025b), and is shown in (2). In it, the word *pē* ‘husband’ is downstepped because the M noun *nōŋ* ‘person’ is positioned at the right edge of the phonological phrase.

- (2) *nōŋ*    *‘pē*    *ǰá*  
 (nōŋ)<sub>ϕ</sub> (pē    ǰá)<sub>ϕ</sub>  
 person husband watch  
 ‘The person watched the husband.’

Downstep likewise occurs if an M follows an HM word at the right edge of the phonological phrase, as in (3).

- (3) *dī*    *‘pē*    *ǰá*  
 (dī)<sub>ϕ</sub> (pē    ǰá)<sub>ϕ</sub>  
 man husband watch  
 ‘The man watched the husband.’

This effect is only conditioned by M or HM tones—if an H word is in the same context, there is no downstep, as in (4).

- (4) *ádámá*    *sú*    *ǰá*  
 (ádámá)<sub>ϕ</sub> (sú    ǰá)<sub>ϕ</sub>  
 Adama father watch  
 ‘Adama watched the father.’

Each of these processes is evidence in favor of the second tone in the HM contour being an M, as it behaves as a level M does in these morphophonological contexts. The higher pitch of the initial H, then, is indicative of a phonetic dissimilatory effect, rather than reflecting a distinct phonological category.

Pitch traces of the three-tone contour tones are shown in Figure 21. Though there is not much phonetic distinction visible between LHL and LHM in the plot, these two tonal categories have distinct phonological behaviors—LHM tones will trigger the same phonological processes that M tones do, as discussed above. Likewise, LHL contour tones trigger L spreading, which is also caused by L or HL tones (Struthers-Young 2025a: p. 115).

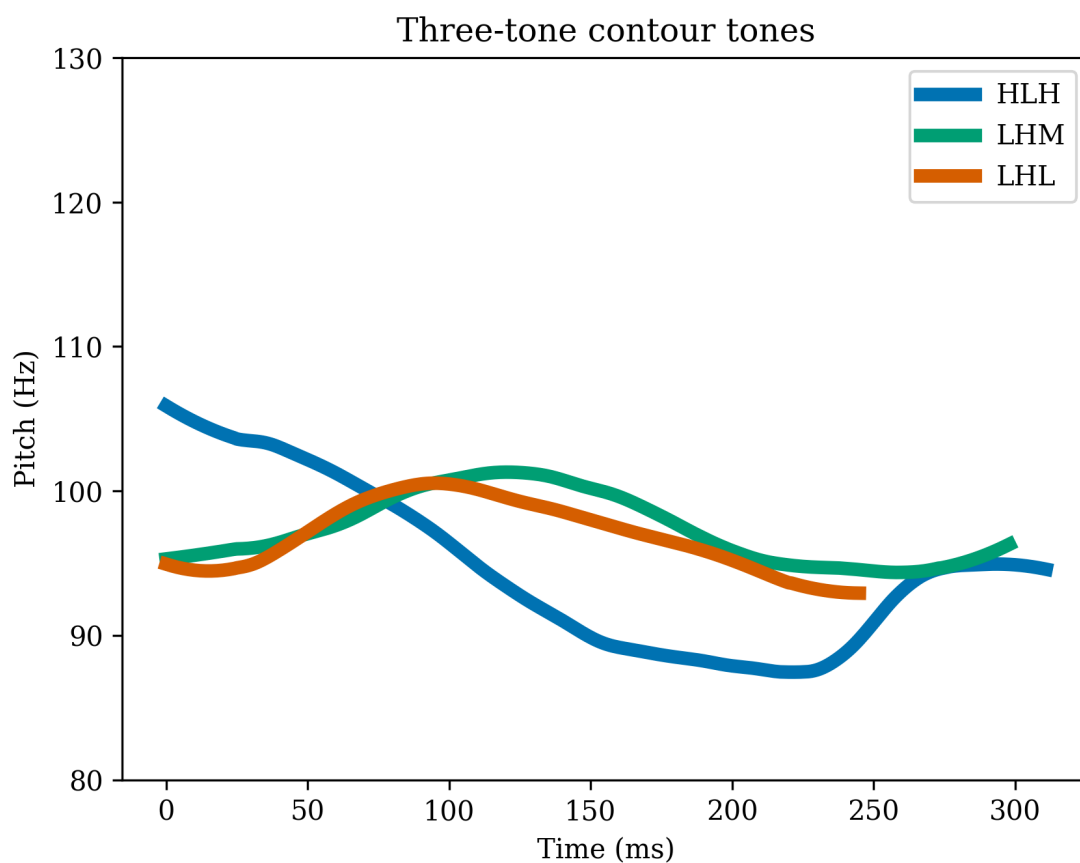


Figure 21: Three-tone contour tones

Figure 22 shows the three-tone contour tones along with the level tones. Syllables with three-tone contour tones are considerably longer than other tonal categories—approximately twice the length of level tones.

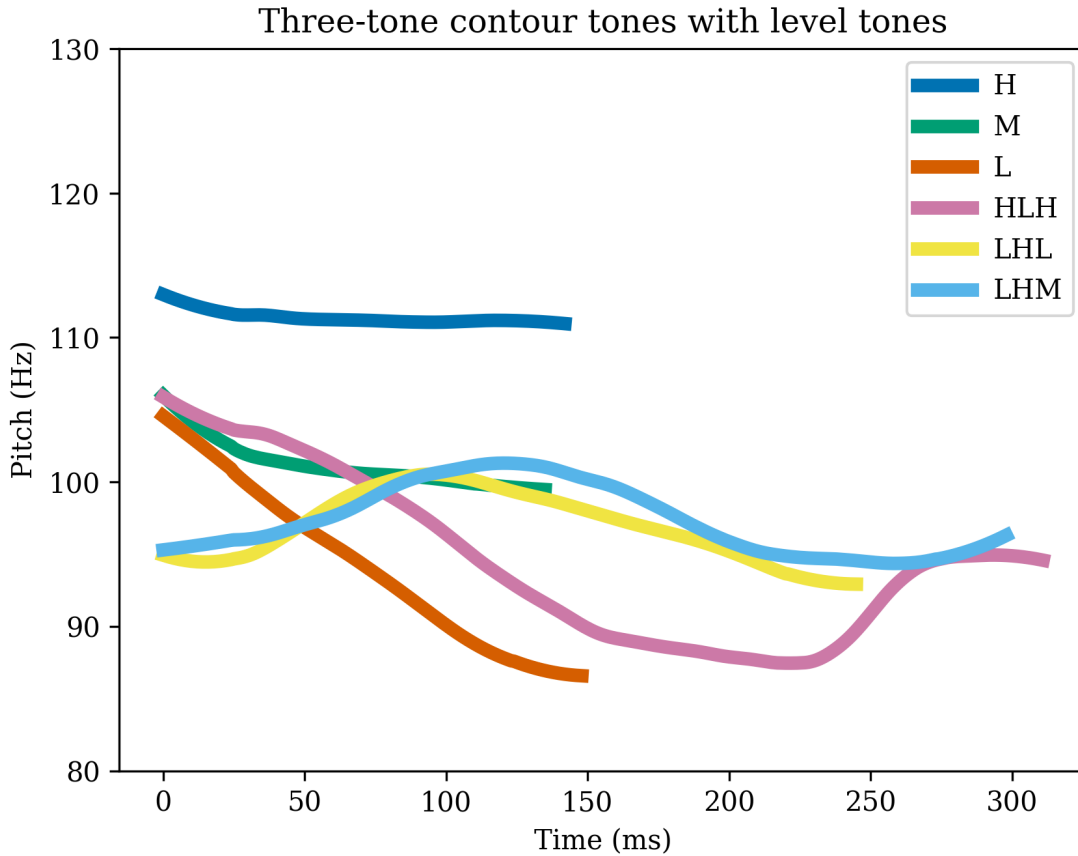


Figure 22: Three-tone contour tones

Downstep is frequent in the language, and both H and M tones can be downstepped. There is automatic downstep, whereby a higher tone is downstepped when preceded by a lower tone, i.e., /M H/ → [M ʰH], /L H/ → [L ʰH], and /L M/ → [L ʰM]. There are also several sources of non-automatic downstep. In addition to the prosodic boundary effect discussed above, there is grammatical tone that indicates that the verb is not preceded directly by an internal argument, i.e., an object or oblique argument. This marker is a floating L tone that either docks onto the verb, resulting in a contour tone, or remains floating, where it triggers downstep—whether docking or downstep occurs is conditioned by multiple factors: 1) when preceded by one class of tense, aspect, and mood markers, the tone docks onto the verb; with another class, the verb is downstepped, and 2) some verbs are always downstepped, regardless of preceding TAMP marker. I call this the *absent preverbal (non-subject) argument* (APVA) marker.

When two different processes that trigger non-automatic downstep target an H tone, the tone is downstepped twice, produced at a pitch lower than a preceding M—typically, a downstepped H is pronounced at or above the level of an M. Double downstep can be seen in the following examples. (5a) shows the effect of the APVA marker, the floating tone that downsteps *pé* COP. When *pé* follows a mid-toned noun like *pē* ‘husband’ in (5), it is downstepped twice—once by the APVA marker, and another time by the prosodic

boundary downstep. I describe and analyze downstep and double downstep in Struthers-Young (2025b).

- (5) a.  $sú$        $^{\circ}pé$   
           $sú$        $^{\circ}pé$   
          father COP.APVA  
          ‘Father is there’
- b.  $p\bar{e}$        $^{\circ\circ}pé$   
       $(p\bar{e})_{\phi}$        $(^{\circ}pé)$   
      husband COP.APVA  
      ‘The husband is there.’

## Phonotactics

The maximal syllable structure of Northern Toussian is CCVC. Of a corpus of 1729 monomorphemic words, half (869/1729; 50.3%) are monosyllabic. There are, however, asymmetries in the typical number of syllables by lexical category. Verbs tend to be monosyllabic, as 75.3% (428/568) of the verbs in the corpus are monosyllabic; 20.2% (115/568) of the verbs are monosyllabic; the rest are polysyllabic. Conversely, nouns tend to be disyllabic. 47.2% (549/1162) of nouns are disyllabic, compared to the 38.0% (442/1162) monosyllabic nouns.

## Onset clusters

In consonant clusters, the first consonant can be stop, fricative, glide, or nasal, and the second consonant is limited to a liquid or glide, one of /l r j w/. Some of the attested consonant clusters are provided in Table 4.

Table 4: Words with consonant clusters

| Cluster   | Word         | Gloss          |
|-----------|--------------|----------------|
| <i>pl</i> | <i>plāk</i>  | ‘bag’          |
| <i>bl</i> | <i>blè</i>   | ‘bamboo’       |
| <i>kl</i> | <i>klō</i>   | ‘millet’       |
| <i>fl</i> | <i>flê</i>   | ‘woman’        |
| <i>ml</i> | <i>mlāk</i>  | ‘noise’        |
| <i>dr</i> | <i>drèː</i>  | ‘tail’         |
| <i>kr</i> | <i>krû</i>   | ‘crocodile’    |
| <i>fr</i> | <i>fribú</i> | ‘beetle’       |
| <i>tj</i> | <i>tjê</i>   | ‘finish’       |
| <i>wj</i> | <i>wjễ</i>  | ‘cord, string’ |
| <i>fw</i> | <i>fwɔ̃</i>  | ‘good’         |
| <i>sw</i> | <i>swā</i>   | ‘field’        |
| <i>mw</i> | <i>mwɔ̃</i>  | ‘rat’          |

The sequence /kw/ freely varies between [kw] and [k<sup>h</sup>p]. I analyze it as a sequence of consonants, rather than a single sound, as it patterns phonotactically more as a consonant cluster. Were it a single sound, clusters like [k<sup>h</sup>pl] or [k<sup>h</sup>pr] might be expected,<sup>9</sup> but are absent. Moreover, since consonant clusters are found in the language, were there the phoneme /k<sup>h</sup>p/, we might expect minimal pairs of /k<sup>h</sup>p/ and /kw/. These are, however, unattested.

## Coda consonants

Possible coda consonants include /n ŋ k r j/. There are a number of restrictions on the vowels that each coda can follow, shown in Table 5. Of these, coda *k* and *j* have the most restrictive distribution, following only /a ā/ and /ɛ ê/, respectively. There are no coda clusters.

Table 5: Possible vowels that codas can follow

| Coda     | Possible nucleus  |
|----------|-------------------|
| <i>k</i> | <i>a, ā</i>       |
| <i>n</i> | <i>i, ɛ, ə, a</i> |
| <i>ŋ</i> | <i>i, ɛ, ɔ, u</i> |
| <i>r</i> | All modal vowels  |
| <i>j</i> | <i>ɛ, ê</i>       |

Examples of words with coda consonants are provided in Table 6.

<sup>9</sup>Such clusters are indeed found in Southern Toussian, which appears to have phonemic labial-velars (Barro, Coulibaly & Wiesmann 2004).

Table 6: Words with coda consonants

| Coda     | Word         | Gloss         |
|----------|--------------|---------------|
| <i>k</i> | <i>pják</i>  | ‘greet’       |
| <i>n</i> | <i>sámān</i> | ‘chicken egg’ |
| <i>ŋ</i> | <i>sóŋ</i>   | ‘sting’       |
| <i>r</i> | <i>swār</i>  | ‘termite’     |
| <i>j</i> | <i>bwēj</i>  | ‘side (N)’    |

Pharyngeal vowels are permitted in syllables with codas, as words with /ɛːj/, /ẽːj/, and /ɛːŋ/ are attested.

## Transcription of a recorded passage

The following passage is a recording of the North Wind and Sun text. The italicized line is a phonetic representation, and the second line is a phonemic one.

*ṁbrī nẽ* ↗  
 ṁbrī nẽ  
 story INDEF

‘A story’

*ṁbrī* <sup>+</sup>pé ↗      *m̃nt̃ŋ nẽm̃nẽ njāː* *nĩŋ*  
 ṁbrī <sup>+</sup>pé ↗      m̃nt̃ŋ nẽm̃nẽ njā    nĩŋ  
 story COP.APVA sun      and      wind on

‘There’s a story about the sun and wind.’

*d̃ m̃nt̃ŋ mĩrĩ*  
 d̃ m̃nt̃ŋ mĩrĩ  
 so sun      think.APVA

‘So the sun thought;’

*b̃ m̃nt̃ŋ pə*      *nsāː*  
 b̃ m̃nt̃ŋ pə      n = sā  
 well sun      PROG IPFV.APVA = think

‘Rather, the sun thinks’

*pâr fâ<sup>s</sup> 'pé pâr ré ñkē njā<sup>s</sup> sē*  
*pâr fâ 'pé pâr ré ñ = kē njā sē*  
 LOG.EMPH power COP.APVA LOG.EMPH at IPFV.APVA = surpass wind with

‘that he is stronger than the wind.’

*njā<sup>s</sup> fáná pá nsā<sup>s</sup> ↗*  
*njā fáná pá n = sã ↗*  
 wind also COP IPFV.APVA = think

‘The wind also thinks’

*fâ<sup>s</sup> 'pé pá ré ñkē mǎntōŋ sē*  
*fâ 'pé pá ré ñ = kē mǎntōŋ sē*  
 power COP.APVA LOG at IPFV = surpass sun with

‘it is stronger than the sun’

*dī nē jǎn kô nǝ fāmǝ páyá sē tá rǝ*  
*dí nē jǎn kô rǝ fāmǝ pák sē tá rǝ*  
 man INDEF rise.APVA voyage in clothes white with 3SG.OBL in

‘A man left for a trip with white clothes on him’

*ñpónō ñpó 'wé*  
*ñ = pó-nō ñ = 'pó 'wé*  
 CON = come-NOM CON = come.APVA see.APVA

‘At their arrival, they come see.’

*mǎntōŋ nǝmǎné mǎntōŋ nǝmǎné njā<sup>s</sup> 'mǎ 'fwó*  
*mǎntōŋ nǝmǎné mǎntōŋ nǝmǎné njā 'pó n = 'fwó*  
 sun and sun and wind PROG.APVA IPFV.APVA = fight.APVA

‘The sun and—the sun and the wind are fighting.’

*ñjǝ nǝŋ mē tē<sup>s</sup> fâ<sup>s</sup> 'pé nǝŋ mē tē<sup>s</sup> rǝ ñkē nǝŋ*  
*ñjǝ nǝŋ mē tē<sup>s</sup> fâ 'pé nǝŋ mē tē<sup>s</sup> rǝ ñ = kē nǝŋ*  
 COMP person DEM owner power COP.APVA person DEM owner in CON = surpass person  
*mē tē<sup>s</sup> sē*  
*mē tē<sup>s</sup> sē*  
 DEM owner with

‘They said that that person is stronger than the other’

*měnná n'jǒ                      bǐ      kà<sup>s</sup>                      'pé                      ɲbàr dàǵǎntò ká      'pé*  
*měnná n = 'jǒ                      bǐ      kà = à                      'pé                      ɲbàr dàkńtò ká      'pé*  
 now    CON = say.APVA    well    3SG.NH = COND    COP.APVA    thus    nothing    NEG    COP.APVA

‘Now, they said, well, if it's like that, that's nothing’

*dǐ ↗ fāmǐ                      ńtò pǒ tǎ                      rǎ ɲdwà                      pá kà                      dǎ                      ɲkǒ*  
*dǐ ↗ fāmǐ                      ńtò pǒ tǎ                      rǎ m = dwà                      pá kà                      dé                      ń = kǒ*  
 so    clothing    REL IS    3SG.H.OBL in    1SG = be.able    LOG    3SG.NH    remove    CON = give

‘So I must be able to remove the clothing that is on him’

*fǎ<sup>s</sup>                      'pé                      tǎ                      sē      tǎ                      tē<sup>s</sup>      rǎ ɲkē*  
*fǎ                      'pé                      tǎ                      sē      tǎ                      tē<sup>s</sup>      rǎ ń = kē*  
 strength    COP.APVA    3SG.H.OBL with    3SG.H.OBL    owner in    CON = surpass

‘He's the strongest.’

*dǐ mǎntǎɲ fáná jǎ                      dàǵǎntò ká      'pé*  
*dǐ mǎntǎɲ fáná jǎ                      dàkńtò ká      'pé*  
 so    sun                      also    say.APVA    nothing    NEG    COP.APVA

‘The sun also said that's nothing’

*ɲǎ<sup>s</sup>      fáná dàǵǎntò ká      'pé*  
*ɲǎ      fáná dàkńtò ká      'pé*  
 wind    also    nothing    NEG    COP.APVA

‘The wind also said that's nothing’

*ɲǎ<sup>s</sup>      pǒ                      tà<sup>s</sup>      tē<sup>s</sup>*  
*ɲǎ      pǒ                      tà      tē<sup>s</sup>*  
 wind    come.APVA    start    take

‘The wind started’

*njã̃<sup>s</sup> p̄ pó ñp̄ pó n̄ pó n̄ pó njã̃<sup>s</sup> p̄ pó lóló*  
*njã̃ p̄ pó ñ = p̄ pó n̄ pó n̄ pó njã̃ p̄ pó lóló*  
 wind IS come CON = IS come CON come CON come wind IS come much

‘When the wind came, it came and came. The wind came a lot’

*d̄ dídjà<sup>s</sup> rì ñtò p̄ jín kô n̄*  
*d̄ dí-djà rì ñtò p̄ jín kô r̄*  
 so man-old DEM.DIST REL IS rise trip in

‘So, the man who left for a trip’

*à ‘pá ñtá jār fjà̃<sup>s</sup>*  
*à ‘pá ñ = t́ jār fjà*  
 3SG.H PROG.APVA CON = 3SG.H.OBL EMPH squeeze

‘He squeezes himself’

*pàsé njã̃<sup>s</sup> lóló ñkã̃<sup>s</sup>*  
*pàsé njã̃ lóló ñ = k̄*  
 because wind much CON = surpass

‘Because there was so much wind’

*k̄m njã̃<sup>s</sup> lõ*  
*k̄m njã̃ lõ*  
 since wind be.much.APVA

‘Because the wind blew hard’

*à ‘pá ñtá jār fjà̃<sup>s</sup>*  
*à ‘pá ñ = t́ jār fjà*  
 3SG.H PROG.APVA CON = 3SG.H.OBL EMPH squeeze

‘he held himself tight,’

*à ‘pá ñtá jār fjà̃<sup>s</sup>*  
*à ‘pá ñ = t́ jār fjà*  
 3SG.H PROG.APVA CON = 3SG.H.OBL EMPH squeeze

‘he held himself tight,’

à      ʔpá      ńtá      jār      fǰǎ̃˥˥  
à      ʔpá      n = tǎ      jār      fǰǎ̃  
3SG.H PROG.APVA CON = 3SG.H.OBL EMPH squeeze

‘he held himself tight.’

nǰǎ̃˥˥ tī      twē pǣj  
nǰǎ̃ tī      twē pǣj  
wind things all do

‘The wind did everything’

nǰǎ̃˥˥ ká kwǎ      nǎ̃ fǎmǎ̃      dǎ      tá      rǎ̃  
nǰǎ̃ ká kwǎ      nǎ̃ fǎmǎ̃      dǎ      tá      rǎ̃  
wind NEG be.able.APVA CON clothing remove 3SG.H.OBL in

‘The wind wasn’t able to remove his clothing.’

ńʔpó      nǰǎ̃˥˥ ńĩr  
ń = ʔpó      nǰǎ̃ ńĩr  
CON = come.APVA wind stop

‘It stopped the wind.’

mǎ̃ntǎ̃ŋ ǰǎ̃      tǎ̃˥      tǎ̃˥  
mǎ̃ntǎ̃ŋ ǰǎ̃      tǎ̃      tǎ̃˥  
sun      rise.APVA beginning take

‘The sun got up and started.’

mǎ̃ntǎ̃ŋ pǎ̃ tǎ̃˥      tǎ̃˥ drǎ̃  
mǎ̃ntǎ̃ŋ pǎ̃ tǎ̃      tǎ̃˥ drǎ̃  
sun      IS beginning grab only

‘The sun only started.’

*ṇp̄p̄ pó n̄ p̄ n̄ p̄*  
*ṇ = p̄ p̄ n̄ p̄ n̄ p̄*  
 CON = IS come CON come.APVA CON come.APVA

‘It came and came and came’

*dídjà<sup>s</sup> n̄<sup>̃</sup> p̄<sup>̃</sup>*  
*dí-djà n̄<sup>̃</sup> p̄<sup>̃</sup>*  
 man-old INDEF COP.APVA

‘There is an old man’

*m̄nt̄ṇ lón̄<sup>̃</sup> p̄j̄ ṇk̄ā<sup>s</sup>*  
*m̄nt̄ṇ ló-n̄<sup>̃</sup> p̄j̄ ṇ = k̄ā*  
 sun much-PL do CON = surpass

‘The sun did the most’

*dídjà<sup>s</sup> rì j̄n̄ t̄ f̄m̄<sup>̃</sup> d̄*  
*dí-djà rì j̄n̄ t̄ f̄m̄<sup>̃</sup> d̄*  
 man-old rise.APVA 3SG.H.OBL clothing remove

‘The man got up and removed his coat.’

*m̄nt̄ṇ á l̄*  
*m̄nt̄ṇ á l̄*  
 sun PST be.much.APVA

‘The sun was a lot’

*k̄m̄ m̄nt̄ṇ p̄<sup>̃</sup> ló*  
*k̄m̄ m̄nt̄ṇ p̄<sup>̃</sup> ló*  
 since sun IS be.much

‘Since the sun is a lot’

*dídjà<sup>s</sup> j̄n̄ t̄ f̄m̄<sup>̃</sup> d̄*  
*dí-djà j̄n̄ t̄ f̄m̄<sup>̃</sup> d̄*  
 man-old rise.APVA 3SG.H.OBL clothing remove

‘The elder got up and removed his coat.’

*dĩ dídjà<sup>s</sup> pō tá fām<sup>h</sup> dě ↗*  
*dĩ dí-djà pō tá fām<sup>h</sup> dě ↗*  
so man-old IS 3SG.H.OBL clothing remove

‘So when the elder removed his coat,’

*njĩ<sup>s</sup> jār wě pâr ká ʼkwó*  
*njĩ jār wě pâr ká ʼkwó*  
wind EMPH see.APVA LOG.EMPH NEG be.able

‘The wind itself saw that it was not able’

*nà dídjà<sup>s</sup> fām<sup>h</sup> dě*  
*nà dí-djà fām<sup>h</sup> dě*  
CON man-old clothing remove

‘to remove the elder’s coat.’

*kōm dídjà<sup>s</sup> jǎ*  
*kōm dí-djà jǎ*  
since man-old say.APVA

‘Since the old man said’

*ñjǒ ñtò à pō kwó pâr fām<sup>h</sup> dě tá tẽ<sup>s</sup>*  
*ñjǒ ñtò à pō kwó pâr fām<sup>h</sup> dě tá tẽ<sup>s</sup>*  
COMP REL DEM.PROX IS be.able LOG.EMPH clothing remove 3SG.H.OBL owner  
*kwǎ*  
*kwǎ*  
be.able.APVA

‘that whoever could remove his clothes wins.’

*njĩ<sup>s</sup> pō pé pé pé ñklō*  
*njĩ pō pé pé pé ñ = klō*  
wind IS IDEO IDEO IDEO CON = tire

‘The wind puffed, puffed, puffed until tired.’

*ṇp̄ṇ pwó lóló ṇp̄ṇ tī twē pēj njā̄s p̄ṇ kwó*  
*ṇ = p̄ṇ pwó lóló ṇ = p̄ṇ tī twē pēj njā̄ p̄ṇ kwó*  
CON = IS come much CON = IS things all do wind IS be.able

‘It did everything that the wind could do.’

*m̄ñnt̄ñ p̄ṇ pó*  
*m̄ñnt̄ñ p̄ṇ pó*  
sun IS come

‘When the sun came,’

*s̄r mār ṇp̄ṇ tó s̄r kwě k̄m m̄ñnt̄ñ j̄ñ*  
*s̄r mār ṇ = p̄ṇ tó s̄r kwě k̄m m̄ñnt̄ñ j̄ñ*  
small only CON = IS again little add since sun on

‘only a little—it added a little more sun.’

*d̄s̄ / dídjà̄s̄ p̄ṇ tá f̄m̄ ðě*  
*d̄s̄ / dí-djà̄ p̄ṇ tá f̄m̄ ðě*  
so man-old IS 3SG.H.OBL clothing remove

‘Then the elder removed his clothing.’

*njā̄s̄ jār s̄*  
*njā̄ jār s̄*  
wind EMPH accept.APVA

‘The wind itself accepted’

*ṇj̄ṇ m̄ñnt̄ñ f̄ā̄s̄ ‘pé m̄ñnt̄ñ p̄ṇ ṇkē p̄ār sē*  
*ṇj̄ṇ m̄ñnt̄ñ f̄ā̄ ‘pé m̄ñnt̄ñ p̄ṇ n = kē p̄ār sē*  
COMP sun power COP.APVA sun PROG IPFV = surpass LOG.EMPH with

‘that the sun is strongest.’

*dṣ̃ ↗ njā̃<sup>s</sup> pṣ̃*  
*dṣ̃ ↗ njā̃ pṣ̃*  
 so wind IS

‘Since when the wind—’

*ṇjṣ̃ mǎntṭṇ̃ nṣ̃ ṇjṣ̃ mǎntṭṇ̃ kwṣ̃ pâr ré*  
*ṇ=jṣ̃ mǎntṭṇ̃ ré ṇjṣ̃ mǎntṭṇ̃ kwṣ̃ pâr ré*  
 CON=say sun to COMP sun be.able LOG.EMPH at

‘it said to the sun, that the sun could win over him.’

*pà twē jēpé mǎntṭṇ̃ kwṣ̃*  
*pà twē jēpé mǎntṭṇ̃ kwṣ̃*  
 3PL.H all see sun be.able

‘They all saw that the sun won.’

*mǎntṭṇ̃ pwṣ̃ nārsē*  
*mǎntṭṇ̃ pwṣ̃ nārsē*  
 sun come.APVA now

‘The sun just came.’

*ṇpṣ̃ kwṣ̃ njā̃<sup>s</sup> re*  
*ṇ=pṣ̃ kwṣ̃ njā̃ ré*  
 CON=IS wind at

‘He could win over the wind.’

*dṣ̃ tîr péj ṇbâr*  
*dṣ̃ tîr péj ṇbâr*  
 so thing do like.that

‘So the thing happened like that’

*ṇjṣ̃ ṇpṣ̃ kà tè<sup>s</sup> tjǎntò ré má kà sà<sup>s</sup> ká tjā*  
*ṇ=jṣ̃ ṇ=pṣ̃ kà tè<sup>s</sup> tjǎntò ré má kà sà ká tjā*  
 1SG=say.APVA 1SG=IS 3SG.NH take REL.LOC at 1SG 3SG.NH put 3SG.NH.OBL place

‘I said that I took it there. I placed it there.’<sup>10</sup>

### Abbreviations

|       |                                              |
|-------|----------------------------------------------|
| AGT   | agentive                                     |
| APVA  | absent preverbal non-subject argument marker |
| COS   | cosubordinate marker                         |
| EMPH  | emphatic                                     |
| FIN   | finite                                       |
| H     | human                                        |
| INDEF | indefinite marker                            |
| IS    | Immediate sequencing marker                  |
| LINK  | linking marker                               |
| LOG   | logophoric pronoun                           |
| NH    | non-human                                    |
| OBL   | oblique                                      |

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<sup>10</sup>This is a common expression used to indicate that a story is finished.

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